

Question				Marking details		Marks Available																					
						AO1	AO2	AO3	Total	Maths	Prac																
5 HT	(a)			<table><tr><td>Wavelength (m)</td><td>Frequency (Hz)</td></tr><tr><td>$> 1 \times 10^{-1}$</td><td>$< 3 \times 10^9$</td></tr><tr><td>$1 \times 10^{-3} - 1 \times 10^{-1}$</td><td>$3 \times 10^9 - 3 \times 10^{11}$</td></tr><tr><td>$7.5 \times 10^{-7} - 1 \times 10^{-3}$</td><td>$3 \times 10^{11} - 4 \times 10^{14}$</td></tr><tr><td>$4 \times 10^{-7} - 7.5 \times 10^{-7}$</td><td>$4 \times 10^{14} - 7.5 \times 10^{14}$</td></tr><tr><td>$1 \times 10^{-8} - 4 \times 10^{-7}$</td><td>$7.5 \times 10^{14} - 3 \times 10^{16}$</td></tr><tr><td>$1 \times 10^{-11} - 1 \times 10^{-8}$</td><td>$3 \times 10^{16} - 3 \times 10^{19}$</td></tr><tr><td>$< 1 \times 10^{-11}$</td><td>$> 3 \times 10^{19}$</td></tr></table>		Wavelength (m)	Frequency (Hz)	$> 1 \times 10^{-1}$	$< 3 \times 10^9$	$1 \times 10^{-3} - 1 \times 10^{-1}$	$3 \times 10^9 - 3 \times 10^{11}$	$7.5 \times 10^{-7} - 1 \times 10^{-3}$	$3 \times 10^{11} - 4 \times 10^{14}$	$4 \times 10^{-7} - 7.5 \times 10^{-7}$	$4 \times 10^{14} - 7.5 \times 10^{14}$	$1 \times 10^{-8} - 4 \times 10^{-7}$	$7.5 \times 10^{14} - 3 \times 10^{16}$	$1 \times 10^{-11} - 1 \times 10^{-8}$	$3 \times 10^{16} - 3 \times 10^{19}$	$< 1 \times 10^{-11}$	$> 3 \times 10^{19}$						
				Wavelength (m)	Frequency (Hz)																						
				$> 1 \times 10^{-1}$	$< 3 \times 10^9$																						
				$1 \times 10^{-3} - 1 \times 10^{-1}$	$3 \times 10^9 - 3 \times 10^{11}$																						
				$7.5 \times 10^{-7} - 1 \times 10^{-3}$	$3 \times 10^{11} - 4 \times 10^{14}$																						
				$4 \times 10^{-7} - 7.5 \times 10^{-7}$	$4 \times 10^{14} - 7.5 \times 10^{14}$																						
				$1 \times 10^{-8} - 4 \times 10^{-7}$	$7.5 \times 10^{14} - 3 \times 10^{16}$																						
				$1 \times 10^{-11} - 1 \times 10^{-8}$	$3 \times 10^{16} - 3 \times 10^{19}$																						
$< 1 \times 10^{-11}$	$> 3 \times 10^{19}$																										
(1) each correct to maximum of 3 marks																											
(b)				<table><tr><td>Region</td><td>Energy (J)</td></tr><tr><td>Radio</td><td>$< 2 \times 10^{-24}$</td></tr><tr><td>Microwave</td><td>2×10^{-24} to 2×10^{-22}</td></tr><tr><td>Visible</td><td>3×10^{-19} to 5×10^{-19}</td></tr><tr><td>X-ray</td><td>2×10^{-17} to 2×10^{-14}</td></tr><tr><td>Gamma ray</td><td>$> 2 \times 10^{-14}$</td></tr></table>		Region	Energy (J)	Radio	$< 2 \times 10^{-24}$	Microwave	2×10^{-24} to 2×10^{-22}	Visible	3×10^{-19} to 5×10^{-19}	X-ray	2×10^{-17} to 2×10^{-14}	Gamma ray	$> 2 \times 10^{-14}$										
				Region	Energy (J)																						
				Radio	$< 2 \times 10^{-24}$																						
				Microwave	2×10^{-24} to 2×10^{-22}																						
				Visible	3×10^{-19} to 5×10^{-19}																						
				X-ray	2×10^{-17} to 2×10^{-14}																						
				Gamma ray	$> 2 \times 10^{-14}$																						
Mark is for correct values for radio and gamma rays																											
Question 5 total		0	3	1	4	4	0																				